



TIME SERIES ANALYSIS

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Some Popular Stochastic Processes



1. White

Noise:

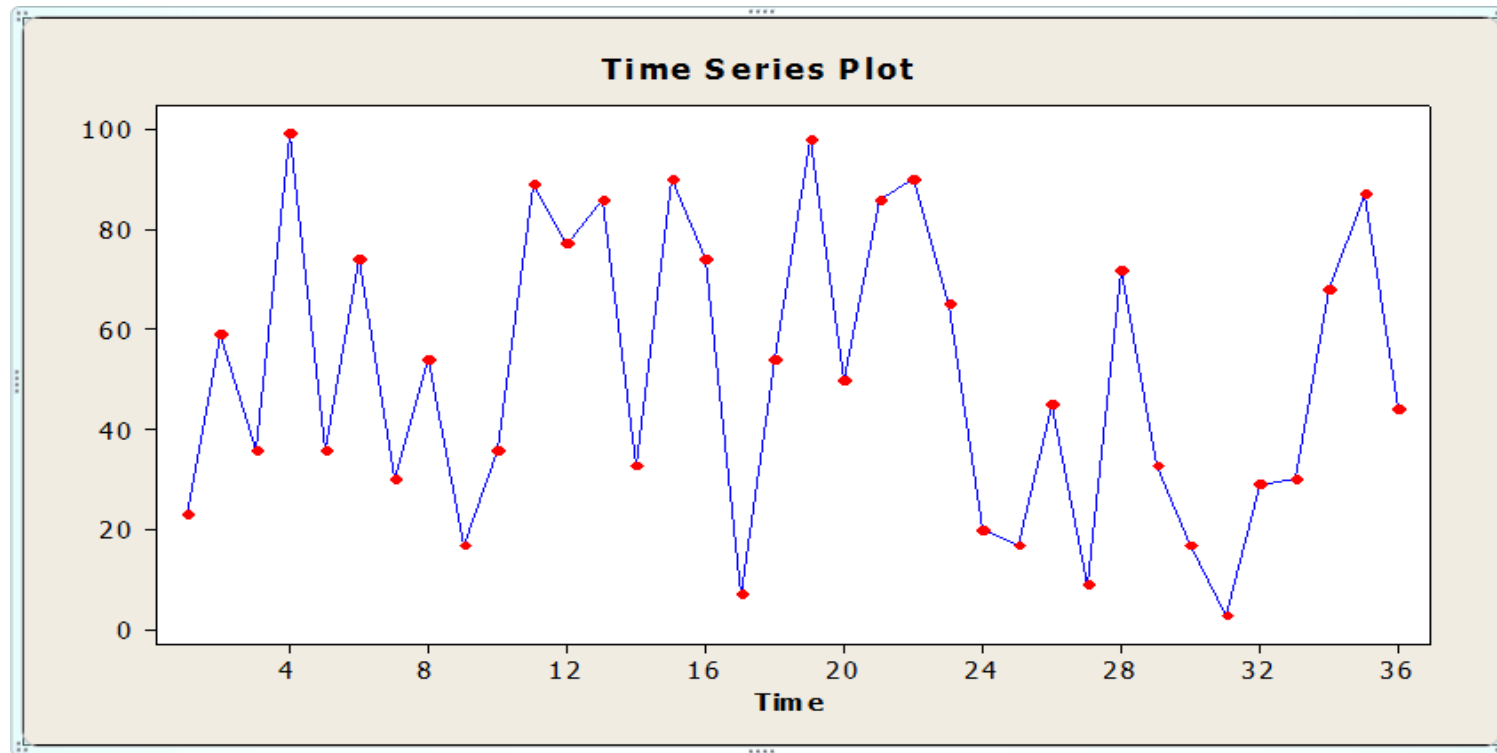


White noise

- This is a purely random process, a sequence of uncorrelated random variables
- Has constant mean and variance
- Also

$$\gamma_k = \text{cov}(y_t, y_{t+k}) = 0, \quad k \neq 0$$
$$\gamma_k = \begin{cases} 1 & k = 0 \\ 0 & k \neq 0 \end{cases}$$

An Illustrative plot of a white noise series





White-noise

- The discrete-time white-noise sequence $\{e[k]\}$ is a set of independent, identically distributed (i.i.d.) values belonging to a stationary stochastic process.
- White-noise sequence exhibits an impulse-like ACF or constant power spectrum
- The ACF of a white-noise sequence is given by:

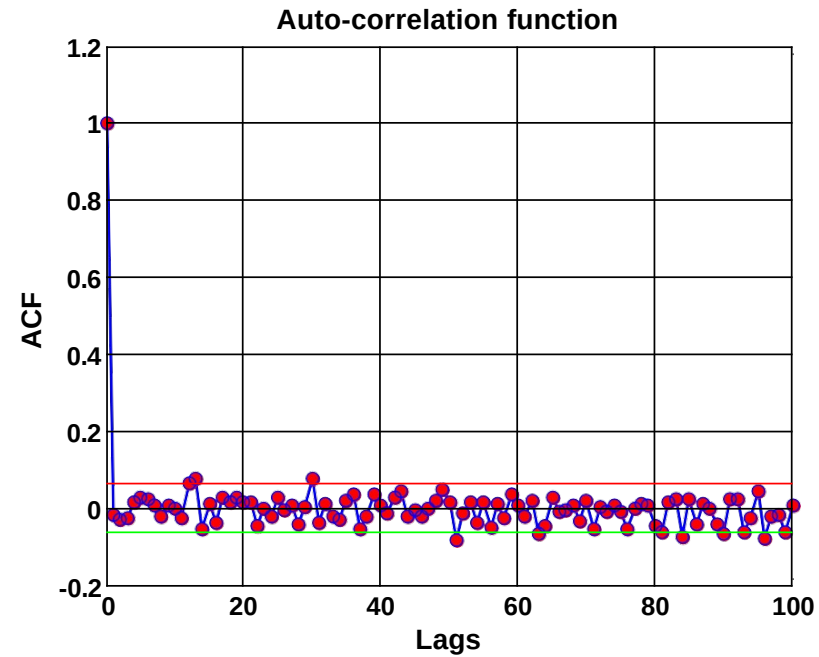
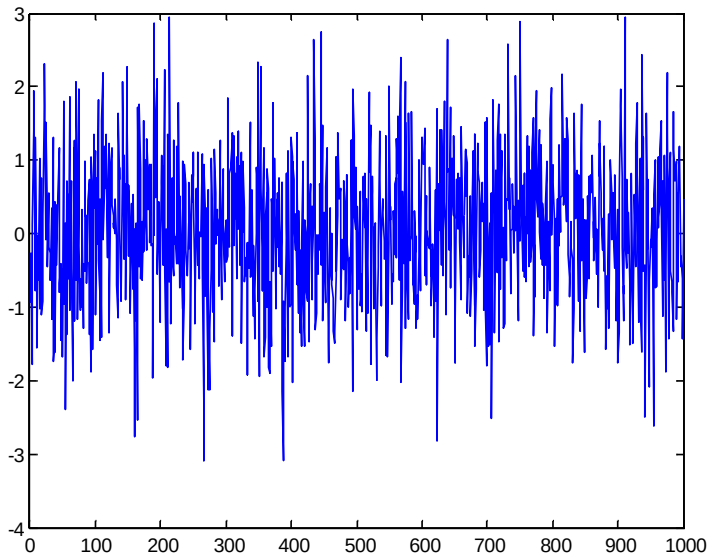
$$\rho_{ee}[l] = \begin{cases} 1 & l = 0 \\ 0 & \text{otherwise} \end{cases}$$

- The power spectrum of white-noise is constant and given by:

$$\Gamma_{ee}(\omega) = \frac{\sigma_e^2}{2\pi} \quad \forall \omega$$

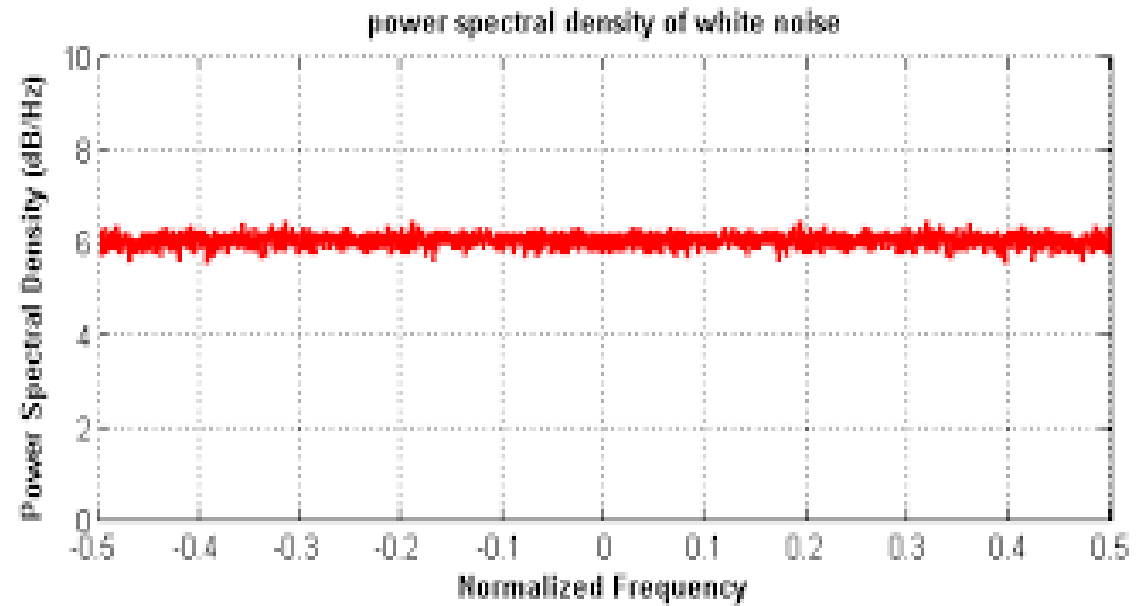
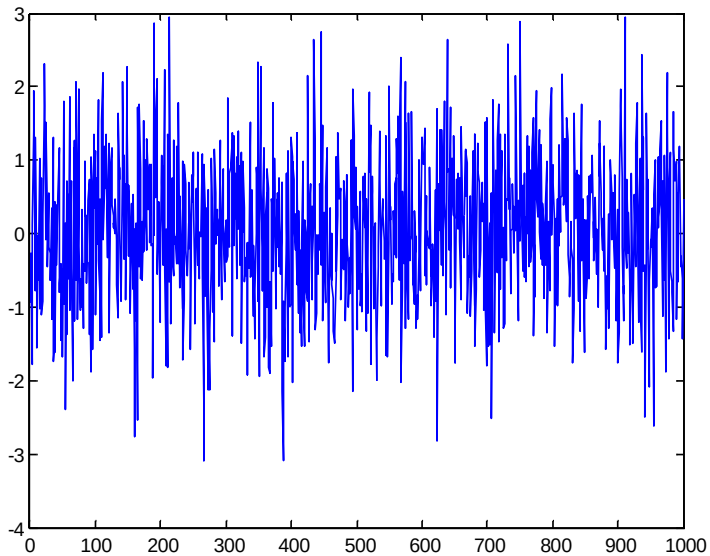
Autocorrelation Function

- ACF of a random data set – zero correlation with itself – indicative of absence of predictability (white data set)



Power Spectral Density

- PSD of a random data set





Wold Decomposition or Wold Representation theorem (without proof):

A zero-mean covariance stationary non-deterministic process can be decomposed as

$$y_t = \sum_{i=0}^{\infty} \theta_i u_{t-i} + \mu_t, \quad (1)$$

where

□ $\theta_0 = 1$ and $\sum_{i=0}^{\infty} |\theta_i|^2 < \infty$,

□ $\{u_t, t \in T\}$ is a purely random process with mean zero and constant variance,

□ μ_t is a purely deterministic process,

□ $\{\theta_i\}$ and $\{u_t, t \in T\}$ are unique.

□ $E(\mu_t u_s) = 0$ for all t and s .



Definition: A zero-mean nondeterministic covariance-stationary process, $\{y_t; t \in T\}$, is called purely nondeterministic if the purely deterministic component $\mu_t = 0$.

Some aspects of Wold Decomposition

- Allows the dynamic evolution of $\{y_t\}$ to be approximated by a linear model.
- The linear model is the only possible representation relating the observed value of y_t to its past evolution of purely random process $u_t, u_{t-1}, \dots$
- Depends on an infinite number of parameters.
- In practice parameters decay rapidly.



Wold Representation as ratio of two finite lag polynomials (without proof):

Under general conditions, the infinite lag polynomial of Wold decomposition can be approximated by the ratio of two finite-lag polynomials:

$$\Psi(B) = \frac{\Theta(B)}{\Phi(B)}$$

where

$$\Phi(B) \equiv (1 - \phi_1 B - \phi_2 B^2 \cdots - \phi_p B^p)$$

$$\Theta(B) \equiv (1 + \theta_1 B + \theta_2 B^2 \cdots + \theta_q B^q).$$



General Linear Process

The following process is called the General Linear Process:

$$y_t = \mu + u_t + \sum_{i=1}^{\infty} \theta_i u_{t-i} \quad (2)$$

Here $\{u_t, t \in T\}$ is a purely random process with mean zero and variance σ_u^2 .

Mean of the process: $E(y_t) = \mu$.



Taking mean $\mu=0$, we can write the process (2) as

$$y_t = \left(1 + \sum_{i=1}^{\infty} \theta_i B^i \right) u_t \quad (3)$$

Let

$$\Theta(B) = \left(1 + \sum_{i=1}^{\infty} \theta_i B^i \right) = \sum_{i=0}^{\infty} \theta_i B^i ; (\theta_0 = 1).$$

Then (3) becomes

$$y_t = \Theta(B)u_t \quad (4)$$



Transfer Function

Let $\{u_t\}$ and $\{y_t\}$ be two time series with

$$y_t = \Theta(B)u_t$$

Then $\Theta(B)$ is also called the *Transfer Function* as it transfers the changes in time series $\{u_t\}$ to time series $\{y_t\}$.

Here the transfer function is

$$\Theta(B) = \left(1 + \sum_{i=1}^{\infty} \theta_i B^i\right) = \sum_{i=0}^{\infty} \theta_i B^i$$



$\theta(B)$ is called an *Impulse Response Function*.

If the input $\{u_t\}$ consists of a unit impulse at time zero ($u_0 = 1$, $u_t = 0 \forall t \neq 0$), then the output (response) is

$$y_t = \sum_{i=0}^{\infty} \theta_i u_{t-i} = \theta_t \forall t = 0, 1, 2, \dots$$

θ -weights give the response when the input is an “impulse”.

Note: In signal processing Discrete Time Unit Impulse Function is defined as $\delta(n) = 1$ if $n = 0$ and $\delta(n) = 0 \forall n \neq 0$.



Linear Filter

$\Theta(B)$ models the interrelationship between input white noise process $\{u_t; t \in T\}$ and output time series $\{y_t; t \in T\}$.

A filter is an operator, given a time series $\{u_t\}$, it maps to a time series $\{y_t\}$.

$\{y_t; t \in T\}$: Linear filtered version of white noise process $\{u_t; t \in T\}$

$\Theta(B) = (1 + \sum_{j=1}^{\infty} \theta_j B^j)$: Linear filter

Linear filter is time invariant (shift invariant)

θ_j : Filter coefficients



Describing the dynamics of a Time Series

- Moving Average (MA) processes
- Autoregressive (AR) processes
- Autoregressive / Moving Average (ARMA) processes
- Autoregressive conditional heteroscedastic (ARCH) processes



2. Moving Average Processes



MOVING AVERAGE PROCESSES

- Suppose you win 1 Dollar if a fair coin shows a head and lose 1 Dollar if it shows tail. Denote the outcome on toss t by a_t .

$$e_t = \begin{cases} 1, & \text{head shows up} \\ -1, & \text{tail shows up} \end{cases}$$

- The average (y_t) winning from the 4 tosses:

$$y_t = \frac{1}{4} e_t + \frac{1}{4} e_{t-1} + \frac{1}{4} e_{t-2} + \frac{1}{4} e_{t-3} \Rightarrow \text{Moving average process}$$



MOVING AVERAGE PROCESSES

- Notice that the observed series (y_t) is autocorrelated even though the generating series e_t is uncorrelated.
- The series (y_t) is the weighted aggregation of some uncorrelated random variables.
- In Economics, the generating series, e_t , is called the *random shock*.
- Random shocks are generally unobserved and are thought to be some unobserved economic activity.



Moving average processes

- The mean and variance are given by

$$E(y_t) = 0, \quad \text{var}(y_t) = \sigma_e^2 \sum_{k=0}^q \theta_k^2$$

Moving average processes

- If the e_t 's are normal then so is the process, and it is then strictly stationary.
- The autocorrelation is

$$\rho_k = \begin{cases} 1 & \text{if } k = 0 \\ \sum_{i=0}^{q-k} \theta_i \theta_{i+k} / \sum_{i=0}^q \theta_i^2 & \text{if } k = 1, 2, \dots, q \\ 0 & \text{if } k > q \\ \rho_{-k} & \text{if } k < 0 \end{cases}$$

The process is weakly stationary because the mean is constant and the covariance does not depend on t .



Moving average processes

- Note the autocorrelation cuts off at lag q
- For the MA(1) process with $\theta_0 = 1$.

$$\rho_k = \begin{cases} 1 & \text{if } k = 0 \\ \theta_1 / 1 + \theta_1^2 & \text{if } k = \pm 1 \\ 0 & \text{otherwise} \end{cases}$$



Moving average processes

- In order to ensure there is a unique MA process for a given acf, we impose the condition of **invertibility**
- This ensures that when the process is written in series form, the series converges
- For the MA(1) process $y_t = e_t + \theta e_{t-1}$, the condition is $|\theta| < 1$



MA(1) and Invertibility

Define

$$\begin{aligned}X_t &= W_t + \theta W_{t-1} \\ &= (1 + \theta B)W_t.\end{aligned}$$

If $|\theta| < 1$, we can write

$$\begin{aligned}(1 + \theta B)^{-1} X_t &= W_t \\ \Leftrightarrow (1 - \theta B + \theta^2 B^2 - \theta^3 B^3 + \dots) X_t &= W_t \\ \Leftrightarrow \sum_{j=0}^{\infty} (-\theta)^j X_{t-j} &= W_t.\end{aligned}$$

That is, we can write W_t as a *causal* function of X_t .

We say that this MA(1) is **invertible**.



MA(1) and Invertibility

$$X_t = W_t + \theta W_{t-1}$$

If $|\theta| > 1$, the sum $\sum_{j=0}^{\infty} (-\theta)^j X_{t-j}$ diverges, but we can write

$$W_{t-1} = -\theta^{-1} W_t + \theta^{-1} X_t.$$

Just like the noncausal AR(1), we can show that

$$W_t = -\sum_{j=1}^{\infty} (-\theta)^{-j} X_{t+j}.$$

That is, we can write W_t as a linear function of X_t , but it is not causal. We say that this MA(1) is not **invertible**.



Invertibility

A linear process $\{X_t\}$ is **invertible** (strictly, an **invertible function of $\{W_t\}$**) if there is a

$$\pi(B) = \pi_0 + \pi_1 B + \pi_2 B^2 + \dots$$

with
$$\sum_{j=0}^{\infty} |\pi_j| < \infty$$

and
$$W_t = \pi(B)X_t.$$



MA(1) and Invertibility

- Invertibility is a property of $\{X_t\}$ **and** $\{W_t\}$.
- Consider the MA(1) process defined by $X_t = \theta(B)W_t$ (with $\theta(B) = 1 + \theta B$):

$X_t = \theta(B)W_t$ is invertible

iff $|\theta| < 1$

iff the root z_1 of the polynomial $\theta(z) = 1 + \theta z$ satisfies $|z_1| > 1$.



Moving average processes

- The general condition for invertibility is that all the roots of the equation $\theta(B) = 0$ lie outside the unit circle (have modulus less than one)



Moving average processes

- For general processes introduce the backward shift operator B .

$$B^j y_t = y_{t-j}$$

- Then the $MA(q)$ process is given by

$$y_t = (\theta_0 + \theta_1 B + \theta_2 B^2 + \cdots + \theta_q B^q) e_t = \theta(B) e_t$$



MA: Stationarity

- Consider an MA(1) process without drift:

$$y_t = e_t + \theta e_{t-1}$$

- It can be shown, regardless of the value of, that

$$\begin{aligned} \text{var}(y_t) &= \sigma_e^2(1 + \theta^2) \\ \text{cov}(y_t y_{t-s}) &= \begin{cases} -\theta \sigma_e^2 & \text{if } s = 1 \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$



MA: Stationarity

- For an MA(2) process

$$y_t = e_t + \theta_1 e_{t-1} + \theta_2 e_{t-2}$$

$$E(y_t) = 0$$

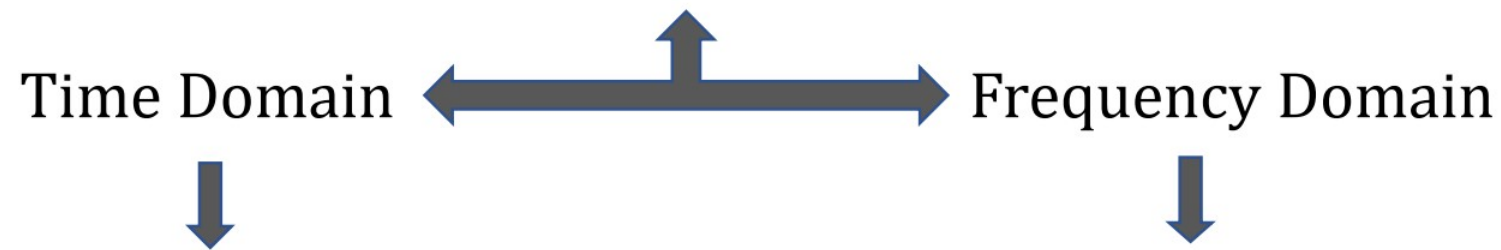
$$\text{var}(y_t) = \sigma_e^2(1 + \theta_1^2 + \theta_2^2)$$

$$\text{cov}(y_t y_{t-s}) = \begin{cases} -\theta_1 \sigma_e^2 & \text{if } s = 1 \\ -\theta_2 \sigma_e^2 & \text{if } s = 2 \\ 0 & \text{otherwise} \end{cases}$$



MA: Stationarity

- In general, MA processes are stationarity regardless of the values of the parameters, but not necessarily “invertible”.
- An MA process is said to be invertible if it can be converted into a stationary AR process of infinite order.
- In order to ensure there is a unique MA process for a given acf, we impose the condition of **invertibility**.
- Therefore, invertibility condition for MA process serves two purposes: (a) it is useful to represent an MA process as an (infinite order) AR process; and (b) it ensures that for a given ACF, there is a unique MA process.



- Based on Wold decomposition
 - Lead to construction of ARMA type parametric models
 - Analysis *wrt* time based on correlation
 - *ACF* plays crucial role
 - Emphasis on how a signal changes over time
- Based on Fourier approximation
 - Lead to construction of weighted function of sine/cosine waves
 - Analysis *wrt* frequencies
 - *Spectrum df* plays crucial role
 - Emphasis on how much of the signal lies within each frequency band



Spectral Density Function of some Stationary Processes:

Purely Random Process: For a purely random process

$$\gamma_k = \begin{cases} \sigma_u^2 & \text{if } k=0 \\ 0 & \text{otherwise} \end{cases}$$

Hence

$$f(\omega) = \frac{\sigma_u^2}{2\pi}; -\pi \leq \omega \leq \pi$$

The spectrum is constant in the range $[-\pi, \pi]$ (like spectrum of white light).

Purely random process is also called *white noise*.

$$\begin{aligned} f(\omega) &= \frac{1}{2\pi} \left[\gamma_0 + 2 \sum_{k=1}^{\infty} \gamma_k \cos \omega k \right] \\ &= \frac{1}{2\pi} \gamma_0 = \frac{\sigma_u^2}{2\pi} \\ f^*(\omega) &= \frac{1}{2\pi}; -\pi \leq \omega \leq \pi \end{aligned}$$



MA(1) Process: $y_t = u_t + \theta u_{t-1}$

$$f(\omega) = \frac{\sigma_y^2}{2\pi} \left[1 + \frac{\theta}{1+\theta^2} \cos \omega \right]$$

$$\text{ACVF: } \gamma_k = \begin{cases} \sigma_y^2 = (1 + \theta^2) \sigma_u^2, & k = 0 \\ \frac{\theta \sigma_y^2}{(1 + \theta^2)}, & k = \pm 1 \\ 0, & \text{otherwise} \end{cases}$$

Power Spectrum:

$$f(\omega) = \frac{\sigma_y^2}{2\pi} \left[1 + \frac{2\theta}{1 + \theta^2} \cos \omega \right]$$

$$= \frac{\sigma_u^2}{2\pi} [1 + \theta^2 + 2\theta \cos \omega]; \quad -\pi \leq \omega \leq \pi$$







Normalized spectral density:

$$f^*(\omega) = \frac{1}{2\pi} \left[1 + \frac{2\theta}{1 + \theta^2} \cos \omega \right]; -\pi \leq \omega \leq \pi$$

- When $\theta > 0$, power is concentrated at low frequencies
- When $\theta < 0$, power is concentrated at high frequencies



When $\theta > 0$, power is concentrated at low frequencies

When $\theta < 0$, power is concentrated at high frequencies

